

Daniel Brown

Senior Data Engineer & BI / ETL Analyst

(978) 731-9708

work.danielpy@gmail.com

Houston, TX

SUMMARY

I am a Senior Data Engineer & BI / ETL Analyst with over 10 years of experience building data-driven solutions that turn raw information into actionable insight. My expertise spans Python, Snowflake, Databricks, and modern ETL/ELT orchestration across hybrid multi-cloud environments (AWS, Azure, GCP). I design and optimize scalable data pipelines, semantic models, and analytics frameworks using Power BI and Tableau—owning enterprise data governance, DAX modeling, and RLS configuration to ensure accuracy and security. Experienced in AI-augmented data platforms, I integrate OpenAI and ML-based anomaly detection into reporting workflows to automate data quality monitoring and accelerate decision-making across business domains.

SKILLS

Languages & Scripting: Python, SQL, PySpark, TypeScript, JavaScript (ES6+), C#, Java, Scala (basic), Bash

Data Engineering & Pipelines: ETL/ELT, Airflow, AWS Glue, Azure Data Factory, Databricks, Spark, Snowflake, BigQuery, Streaming Pipelines (Kafka, Kinesis)

Cloud Platforms: AWS (S3, Glue, Lambda, Redshift, Athena), GCP (BigQuery, Pub/Sub, Cloud Functions), Azure (Synapse, Data Lake, Functions), Hybrid Multi-Cloud

BI & Analytics: Power BI (Dataflows, DAX, RLS, Certified Datasets, Governance, Semantic Layer Ownership), Tableau, Looker

AI & ML: OpenAI / Azure OpenAI, LangChain, ML-based Anomaly Detection, Data Quality Scoring, RAG Pipelines

Architecture & Design: Data Lakehouse, Event-Driven Architecture, Microservices, Data Mesh Concepts

DevOps & Automation: Docker, Kubernetes (EKS/GKE), Terraform, Helm, Jenkins, GitHub Actions, CI/CD for Data Pipelines

Security & Governance: OAuth 2.0, RBAC, Data Masking, Encryption, HIPAA/PCI Compliance, Data Lineage

Monitoring & Observability: Prometheus, Grafana, ELK Stack, CloudWatch, Datadog

Soft Skills: Agile Delivery, Cross-Functional Collaboration, Mentoring, Stakeholder Communication, Data Strategy Alignment

PROFESSIONAL EXPERIENCE

Senior Data Engineer / BI & ETL Lead

Tata Consultancy Service (TCS)

10/2022 - Present

Environment: Python, PySpark, Snowflake, Databricks, Power BI, Tableau, AWS + GCP Hybrid Multi-Cloud

- Designed and maintained multi-cloud data pipelines using Python, Airflow, and AWS Glue to process billions of rows daily with <1% error rate.
- Built and owned enterprise Power BI semantic models with DAX, RLS, and governed datasets, improving reporting consistency and KPI alignment across 12 departments.
- Engineered high-performance ETL pipelines in Databricks and Snowflake to deliver real-time analytics with 40% faster load times.
- Integrated OpenAI-powered anomaly detection and data quality scoring into ETL validation layers to automate error tracking.
- Developed Tableau dashboards and Power BI reports supporting cross-cloud financial, operational, and customer insights.
- Implemented serverless compute using AWS Lambda and GCP Cloud Functions for automated metadata refresh and incremental ingestion.
- Built data lakehouse framework on S3 + Databricks Delta for unified data access and versioned storage.
- Orchestrated CI/CD automation for data pipelines using Terraform, Jenkins, and GitHub Actions, cutting deployment times by 60%.
- Created governed enterprise data catalog with lineage, sensitivity classification, and ownership metadata.
- Reduced query cost and runtime in Snowflake by 35% through clustering and materialized caching optimization.
- Mentored five junior data engineers and BI analysts in Python scripting, data modeling, and governance best practices.

Senior Data Engineer

Accenture

06/2016 - 09/2022

Environment: Python, SQL, Snowflake, Power BI, Tableau, Azure Primary (AWS selective utilities)

- Developed and maintained ETL pipelines integrating financial data from SQL Server, PostgreSQL, and RESTful APIs into Snowflake for enterprise analytics.
- Built Power BI and Tableau dashboards supporting real-time financial insights and automated performance reporting.
- Led migration from on-prem SSIS pipelines to Azure Data Factory, achieving 99.9% data delivery reliability.
- Designed enterprise Power BI semantic layer and governance model, implementing RLS, DAX optimizations, and certified datasets.
- Created AI-enhanced data classification workflow using Azure OpenAI to automate metadata tagging and column-level PII detection.
- Built Python-based ETL scripts for incremental data loads and API integrations with third-party financial

systems.

- Managed SQL performance tuning, index optimization, and materialized view creation to improve query speed by 45%.
- Implemented monitoring dashboards in Grafana and Prometheus to track ETL latency and pipeline health.
- Deployed Databricks jobs for Spark-based data transformations and distributed computation.
- Partnered with business analysts and leadership teams to define metrics, data sources, and reporting KPIs.

Data Engineer

Amazon

09/2015 - 05/2016

Environment: Python, Flask, AWS, Redshift, Kafka, Power BI

- Developed Python-based ETL jobs to consolidate operational data into AWS Redshift, supporting enterprise analytics across multiple geographies.
- Built APIs and data connectors for ingestion of financial data into Power BI for executive dashboards.
- Engineered Kafka-based data streams for real-time order analytics and forecasting models.
- Implemented Flask microservices for data access layers serving analytics applications.
- Designed Power BI datasets and reports with DAX measures for logistics, sales, and inventory analysis.
- Optimized ETL workflows by containerizing jobs in Docker and orchestrating through AWS ECS.
- Created internal data quality validation scripts using Python and Pandas for pipeline testing.
- Collaborated with Data Science teams to deploy predictive ML models into production analytics dashboards.

EDUCATION

Stony Brook University

Master of Business Administration (MBA)

2018 – 2020

Stony Brook University

Bachelor of Computer Science

2011 – 2015